# Secured Cloud Authentication using Quantum Cryptography

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***Abstract*—Cloud computing have become the budding engine of current information technology (IT) as a result of the rapid development of IT, security protection has become more and more important for cloud computing. The theft of identity authentication in sequence remains a serious hazard to the protection of cloud computing, in this process, attackers infringe into cloud computing services through identity certification information, thereby threatening the security of data from multiple perspectives. Most types of attack can cause the classified information be in the hand of unauthorized party. This situation will lead to data disclosure and the user feel insecure for using the services offered by cloud infrastructure. In this work a novel protocol is proposed to secured cloud authentication using quantum cryptography.**

***Keywords—Authentication; Cloud Authentication; Quantum key distribution; QKD Protocol.***

## I. INTRODUCTION

The purpose of the quantum cryptography field which combines quantum theory with information theory is to use the laws of physics to provide secure information exchange, when compared to the classical methods based on complexity assumption. QKD protocols became important due to various technological advances which allow their implementation in the laboratory. To ensure the safety measures of quantum key in real time applications both Quantum Key Distribution and Quantum Key Authentication is important.

Quantum authentication [1] methods are fairly useful. Not only can they be used to consistently send quantum data all the way through an untrusted channel, they also offer rewards for sending classical information. Since this scheme both encrypts and authenticates quantum information, clearly it also encrypts and authenticates classical data transfer via the protocol.This much could be done with the sole purpose classical techniques, but the quantum authentication protocol offers an extra property impossible in the classical world: the adversary Eve is not capable to copy the message throughout transmission. That is, a quantum authentication scheme also provides uncloneable encryption.

Several quantum validation methods have been proposed. Duseketal proposed a scheme which is a amalgamation of classical identification method and quantum key distribution. Zeng's protocol is by means of EPR pairs as the first authentication keys and using classical keys circulated in quantum key distribution procedure. Later, two new validation protocols using entanglement and catalysis were proposed by Barnum and Jensen et al respectively in that various parties in communication have earlier shared catalyst. The verifier sends the challenge which is a half part of some entangled state to the identifier, then they can transfer this entangled state to a special catalyst state by local quantum operation and classical communication (LQCC) with the catalyst.As this task cannot be proficient by LQCC without the catalyst, the verifier can authenticate identifier by measuring the state after the identifier sends it back.

Quantum authentication is superior than classical authentication [2] in efficient ways. The quantum no-cloning theorem doesn't allow an eavesdropper to access the identification token. This makes the quantum authentication capable for theft detection as well as the quantumkey distribution gives protection from the eavesdropping. The quantum no-cloning theorem also supports the user without his/her interference he/she can restricts other authorized persons from sharing tokens.

## II. RELATED WORK

Zuriati Ahmad Zukarnain, and Roszelinda Khalid[3] are introducing a new idea that can be adopt in multiparty communication committed for cloud infrastructure. There is a budding vulnerability that someone will interrupt the data being read, masquerade as one of your applications, and fill your system with counterfeit data. To survive with this problem, they use the quantum key distribution to create an encrypted channel between the server and the client. Then the data transaction will send via public channel. From the result they can see there is a significant in terms of error rate. They hope that with the proposed scheme it can be applied on any

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quantum key distribution protocol and able to authenticate the legitimate users.

Geeta Sharma & Sheetal Kalra[4] The idea of employing QKD in cloud computing is novel and meets all vital security necessities. The formal endorsement is performed using AVISPA tool, which confirms its safety measures in the presence of a prospective intruder. To substantiate applicability of Quantumkey Distribution in cloud computing, a practical long-distance entanglement-based QKD experimentation has been proposed. This test confirms successful creation of shifted keys over distance of 100 km of optical fiber with a key rate of 4.11 bit/s and an error rate of 9.21 %. Implementations of quantum communiqué using existing network structure are concrete the way for the commercialization of quantum technology.

YoshitoKanamori, Seong-Moo Yoo, Don A. Gregory and Frederick T. Sheldon[5] have introduced a two-way authentication protocol for a simple authentication case (our multi-party authentication protocol will be discussed in a future paper). To hide conveyed data from unapproved users, this protocol uses quantum super positioned states as a replacement for quantum entangled states (related to other quantum authentication protocols). Remember, it is a challenging task to authenticate a specific user (the most common use of authentication protocols) within a group of many using quantum entangled states. Their protocol works fine under the statement that both parties already share a secret key (Ka). Furthermore, they showed that the superposition states can be apprehended using modern technologies (e.g., linear polarizers and Faraday rotators).

Yumin Dong, Shufen Xiao, Hongyang Ma and Libo Chen[6] projected a model to provide security in cloud computing relied on quantum authentication to thwart attackers from intruding into mature cloud services through information and from damaging data in cloud computing. They introduced the principle of quantum authentication in detail and described the authentication process. The results of this study proved that any entity who attempts to steal the identity of the three elements (i.e., DUser, CSP, and DOwner) of a cloud model to damage data in cloud computing would be discovered.

Howard Barnum, Claude Crepeau, Daniel Gottesman, Adam Smith, and Alain Tap[7] examined several characteristics of the problem of authenticating quantum messages. They have shown the security of a large class of private-key quantum authentication schemes, and presented a particular highly efficient scheme from that class. One feature of the scheme is that it totally encrypts the message, and they show that this is a essential feature of any quantum message authentication code: if any observer can learn a substantial amount of information about the authenticated state, that observer also has a good chance of successfully changing thestate without being identified. Authentication of quantum states in a public key context, and shown that while authentication is possible with public keys, digitally signing quantum states is never possible, even when only computational security is required. The necessity for encryption is rather surprising, given that classical messages can be authenticated without encrypting them. The difference can be understood as a complementarily feature of quantum mechanics: authenticating a message in one basis requires encrypting it in the complementary Fourier-transformed basis. This is essentially another realization of the principle that measuring data in one basis disturbs it in any complementary basis. For classical messages, therefore, encryption is not required: only one basis is relevant.

## III. PROPOSED WORK

The proposed authentication schema is developed based quantum secured direct communication protocol for cloud server authentication, which involves in the reordering of the entanglements where it is assumed that the imposters can alter the key shared through the quantum channel.

*Protocol design*

It is assumed that the proposed authentication scheme for cloud server includes mainly two entities specifically the service provider and service requestor. Initially the service requestor requests the access permissions to the cloud server, whereas the quantum based authentication is exemplified as follows:

*Registration phase:* The authentication parameter is enlisted within the cloud server upon the successful registration of the service requestor

*Communication Phase:* The service requestor initiates the request for the access permissions to the cloud server

*Authentication Phase:* Identity based authentication will be taken place between the mutual parties

### *A. Registration Phase:*

To establish the communication channel with the cloud server (CS), the service requestor transmits the unique requestor id (IDr) to the Cloud Server where in response it generates password (Pwd) for the requestor

Step 1: CSr transmits the unique requestor id (IDr) to CS
Step 2: CS Computes Authentication parameters (wi, xi, pwd)
Step 3: CS stores computed values along with IDr
Step 4: CS transmits pwd to CSr

### *B. Communication Phase:*

To enable the communication between the cloud service provider and service requestor initially the requestor should provide the authentication parameter Pwd to receive the authenticated permissions from the cloud server.

Here in this phase the check points are evaluated in two cases:

Case 1: Unauthorized requestor trying to access the permissions of the cloud server

Step 1: CSr sends IDr and pwd to CS
Step 2: CS verifies If #(IDr ^Pwd) = $X_i$
Access Granted
Else
Request Denied

Case 2: An Imposter trying to alter or identify the ' Pwd ' Communicated by the 'CSr'. In these type of cases the process of two level authentication takes place using short shared key and an Identification group key $I_{GK}$, where the $I_{GK}$ is comprised of two parameters (y, $I_B$): y is a positive integer and $I_B$ is the Identification Block

| $I_{GK1}$ | $I_{GK2}$ | $I_{GK3}$ | $I_{GK4}$ |
|---|---|---|---|
| y, $I_B${,} | y, $I_B${,} | y, $I_B${,} | y, $I_B${,} |

If y is an even number, then horizontal-vertical polarization is applied for encoding Qubits. Otherwise, encode in diagonal polarization for odd y value. Secondly, Discard bit (Db) function discards the first and last bit and utilize the Identification block to retrieve the $I_{GK.}$

Step 1. CSr (authentication key or session key) to CS (*Public Ch*
Step 2. CS sends erroneous qubits to CSr (*Quantum channel*)
Step 3. CSr computes the qubits, and then sends Qerr value to CS (*P*
Step 4. CS compares Qerr value,
if correct, and then connection is established
. else , CS terminates the operation (*Public Channel*)

*C. Authentication Phase:*

**Step 1.** CS generates n pair of EPR pairs.
**Step 2.** CS forms sequence KU=|qj/) by taking initial qubit from generated EPR pairs.
**Step 3.** CS performs Hadamard operation (H) on the photons of KU corresponding to the bit value "1" or "0" of KU and generates KU.
**Step 4.** Ui retrieves Kn from Ky using the pre-shared key Pi.
**Step 5.**From the recovered sequence KU, CS selects 2n qubits as check photons and 2n target qubits. The generated check photons are also engaged as control qubits CS executes the Controlled- NOT(CNOT) process on the control and target photons.
**Step 6.**Depending on the bit value of Pwd, CS executes Hadamard operation H on the check and target qubits.
**Step 7.**Using a public channel, CS declares the positions of the check photons to the user.
**Step 8.**According to the value of pre-shared key Pwd, the user performs Hadamard operation on the corresponding check photons.
**Step 9.**Based on the bit value of Pwd, CS performs following operation:
i. If Pwd = 0, then CS executes CNOT operation on the check and the ancillary photons again.
ii. Else CS employs Z basis to measure the ancillary photon. According to the obtained result |0) or |1)}, CS calculates H on the check photons

## IV. EXPERIMENTAL RESULTS

In this work to overcome the confinements of established key dissemination, quantum enter dispersion is utilized as a part of E-Governance applications. Here QKD test system results are broke down to convey the key utilizing quantum key circulation. The portrayal of stages in QKD Simulator is as per the following:

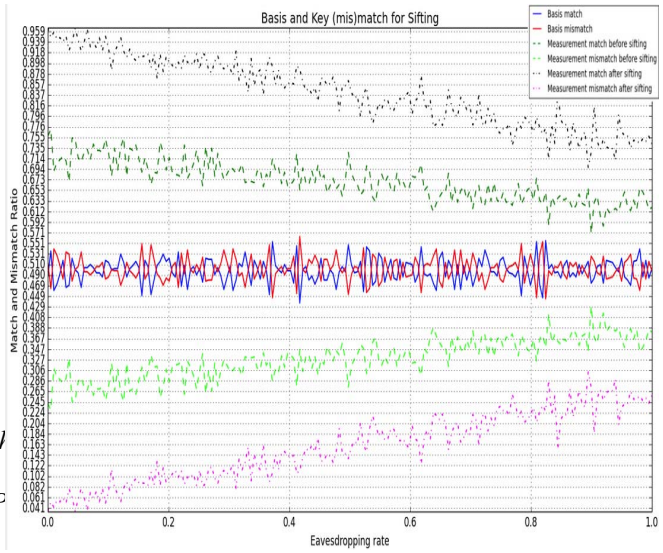


Fig. 1. QKD Sifting Plot

The Fig. 1 shows QKD Sifting Plot generated by QKD simulator. Here matching and mismatching of basis and key, before and after sifting are presented.

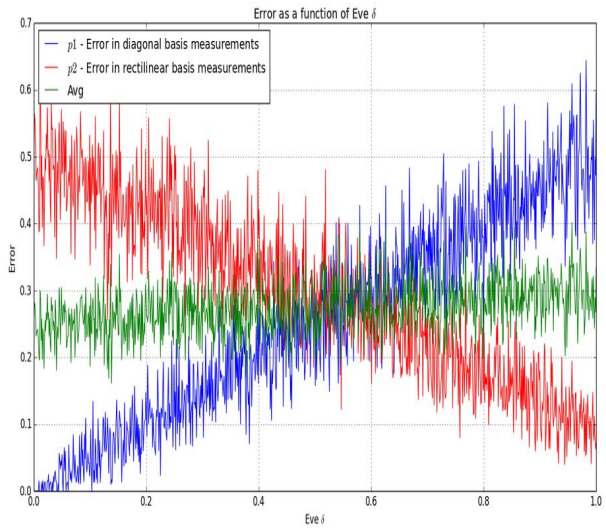


Fig. 2. QKD Biased Error Estimation Plot

The Fig. 2 shows QKD Biased Error Estimation Plot generated by QKD simulator. Here probability of error measurement in diagonal, rectilinear basis and average are presented

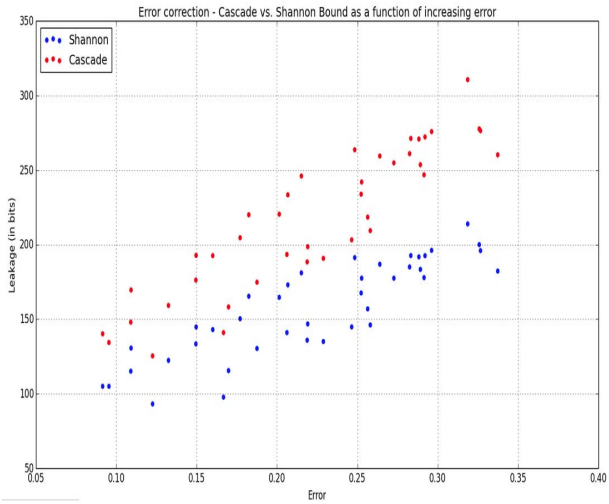


Fig. 3. QKD Shannon Bound vs. Error Correction Plot

The Fig. 3 shows QKD Shannon Bound vs. Error Correction Plot. Here Shannon bound and error correction-cascade as a function changes when leakage changes.

Like the above graphs mentioned in Fig. 1, Fig. 2 and Fig. 3 for each parameter changes simulator generates three graphs like as above. By analyzing all the graphs the following are the statistics of the results.

TABLE I. EAVESDROPPING RATE CHANGES AND ALICE/BOB BASIS SELECTION, EVE BASIS SELECTION KEPT IDLE

| Initial number of qubits | Eavesdropping rate | Information leakage (Total number of disclosed bits) | Key length before error correction | Bit error probability | Bits leaked during error correction | Shannon bound for leakage | Final key length | Estimated error |
|---|---|---|---|---|---|---|---|---|
| 500 | 0.1 | 83 | 376 | 0.0133 | 51 | 39 | 273 | 0 |
| 500 | 0.2 | 85 | 369 | 0.0136 | 53 | 39 | 264 | 0 |
| 500 | 0.3 | 104 | 378 | 0.0212 | 72 | 56 | 254 | 0 |
| 500 | 0.4 | 80 | 370 | 0.0108 | 48 | 32 | 270 | 0 |
| 500 | 0.5 | 169 | 358 | 0.0531 | 137 | 108 | 169 | 0.0256 |
| 500 | 0.6 | 169 | 363 | 0.0523 | 137 | 108 | 174 | 0.0256 |
| 500 | 0.7 | 119 | 373 | 0.0295 | 87 | 72 | 234 | 0 |
| 500 | 0.8 | 159 | 369 | 0.0434 | 127 | 96 | 190 | 0 |
| 500 | 0.9 | 183 | 382 | 0.0602 | 151 | 126 | 179 | 0.0488 |

The table1 shows the how information leakage, key length before error correction, bit error probability, bits leaked during error correction, Shannon bound for leakage, final key length and estimated error varies when Eavesdropping rate changes from 0.1 to 0.9 and remaining parameters kept constant (Alice/Bob basis selection=0.1, Eve basis selection=0.1, Eavesdropping=1, Error estimation sampling rate=0.1, Biased error estimation=1, Error tolerance=0.1). The below figure show the graphical representation of the above table.

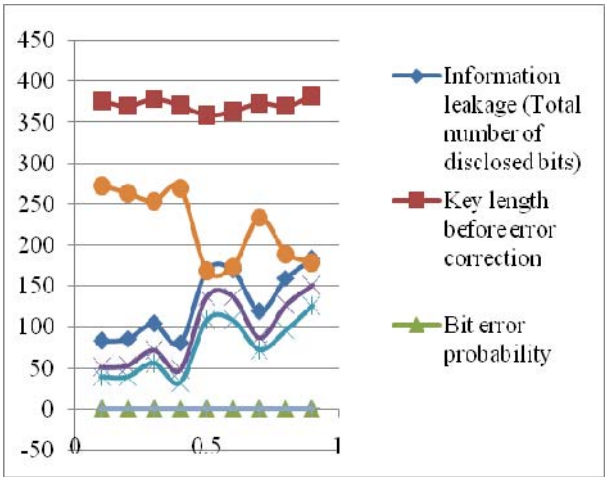


Fig. 4. Eavesdropping rate changes and Alice/Bob basis selection, Eve basis selection kept idle

## V. CONCLUSION

In this work a novel framework for secured cloud authentication using quantum cryptography. The objective of the proposed model is to authenticate the users in cloud using quantum cryptography. For the experimental analysis the BB84 protocol is simulated using QKD simulator that establishes a secured quantum channel. Proposed protocol is evaluated with the probability of 0.1 - 0.9 eavesdropping rate with 500 qubits each and the experimental results specify that required message is decrypted with optimum key length and provides high level confidentiality and authentication when compared with the existing and traditional cryptographic methods for data security and authentication.

## REFERENCES

[1] Zhang X L. One-way quantum identity authentication based on public key. Chinese Sci Bull, 2009, 54: 2018-2021, doi: 10.1007/s11434-009-0350-9.

[2] Bao-Sen Shi, Jian Li, Jin-Ming Liu, Xiao-Feng Fan, Guang-Can Guo "Quantum key distribution and quantum authentication based on entangled state" Physics Letters A 281 (2001) 83–87.

[3] Zuriati Ahmad Zukarnain, and Roszelinda Khalid "Quantum Key Distribution Approach for Cloud Authentication:Enhance Tight Finite Key" ICSIS-2014, pp. 28-33

[4] Geeta Sharma & Sheetal Kalra "Identity based secure authentication scheme based on quantum key distribution for cloud computing" Peer-to-Peer Netw. Appl. DOI 10.1007/s12083-016-0528-2

[5] Y. Kanamori, Seong-Moo Yoo, D. A. Gregory and F. T. Sheldon, "On quantum authentication protocols," *GLOBECOM '05. IEEE Global Telecommunications Conference, 2005.* pp. 1650-1654

[6] Yumin Dong, Shufen Xiao, Hongyang Ma and Libo Chen "Research on Quantum Authentication Methods for the Secure Access Control Among Three Elements of Cloud Computing" Int J Theor Phys DOI 10.1007/s10773-016-3132-6.

[7] Howard Barnum, Claude Crépeau, Daniel Gottesman, Adam Smith, and Alain Tapp "Authentication of Quantum Messages"